

A stringed musical instrument produces sound through the vibrations of strings fixed at both ends. If the velocity at which waves propagate through each string is known, what information is needed in order to determine the fundamental frequency of each string?

- ☐ A. String bulk modulus [4%]
- ✓ ☒ B. String length [87%]
- ☐ C. String density [3%]
- ☐ D. String cross-sectional area [4%]

Correct

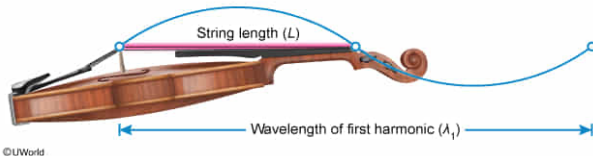
86% answered correctly

Explanation:

Fundamental frequency of a string fixed at both ends

$$f_1 = \frac{v}{\lambda_1} \rightarrow \begin{array}{l} \text{Harmonic wavelengths} \\ \lambda_n = \frac{2L}{n} \\ (n = 1, 2, 3, \dots) \end{array}$$

- f_1 Fundamental frequency (frequency of first harmonic)
- v Wave speed
- λ_1 Wavelength of first harmonic



Standing waves are stationary waves that can be produced within strings, **pipes**, and other common objects. For example, perturbing a string fixed at both ends simultaneously produces multiple standing waves (numbered $n = 1, 2, 3$) known as **harmonics**.

The frequency f of the first harmonic ($n = 1$) is called the **fundamental frequency** (f_1), the lowest-frequency standing wave that can be produced within an object. The given wave frequency of the first harmonic is related to **velocity** (v) and to **wavelength** (λ_1) by the expression

$$f_1 = \frac{v}{\lambda_1}$$

A distinguishing feature of standing waves is that standing wave characteristics are closely related to the dimensions of the objects in which they are produced. Accordingly, the wavelength (λ_n) of each harmonic in a string fixed at both ends is proportional to the **string length** (L):

$$\lambda_n = \frac{2L}{n}$$

Using the above equation, the wavelength of the first harmonic ($n = 1$) may be expressed as

$$\lambda_1 = 2L$$

Substituting this expression for the wavelength of the first harmonic into the equation relating frequency to velocity and wavelength gives

$$f_1 = \frac{v}{2L}$$

Therefore, when provided with wave velocity, only the string length would be needed to determine the fundamental frequency.

(Choices A & C) The **bulk modulus (B)** refers to the compressibility of a substance. In combination with density (ρ), bulk modulus influences the velocity of **sound waves** through fluids (eg, the atmosphere). However, bulk modulus and density are both unrelated to standing waves (ie, those observed in strings and pipes).

(Choice D) The cross-sectional area of a string is unrelated to the wavelength, frequency, or velocity of standing waves that may be produced within that string.

Educational objective:

The fundamental frequency of a wave is directly proportional to wave speed and inversely proportional to wavelength. For a string fixed at both ends, the wavelength of each harmonic is proportional to the string length.

Time spent:0 sec
Subject:
Physics

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System:
Light and Sound